



PROJECT PROPOSAL

Voting & Election Performance Intelligence Dashboard

An Interactive Microsoft Excel Business Intelligence Portfolio Project

Project Type	Business Intelligence / Data Analytics / Data Visualization
Primary Platform	Microsoft Excel (.xlsm)
Design Direction	Application-style dashboard with UI/UX principles
Geographic Scope	Nigeria — 36 States + FCT
Election Scope	INEC-conducted Federal and State elections
Historical Scope	Five most recent completed general-election cycles: 2007, 2011, 2015, 2019 and 2023
Core Questions	Who is voting? Who is winning? What potential irregularity indicators exist?
Project Status	Planning / Dataset Architecture
Author	Wahab Sodiq — Data Analyst Team Leader Business Intelligence & Data Visualization
Portfolio Position	Third Excel analytics/dashboard project

Prepared by Wahab Sodiq
August 2026

Table of Contents

1. Executive Summary.....	3
2. Background and Rationale.....	3
3. Project Scope.....	4
4. Project Objectives.....	4
5. Core Analytical Questions.....	5
6. Election Taxonomy and Data Model.....	6
7. Dataset Architecture.....	6
8. Core Data Dictionary.....	7
9. KPI Framework.....	8
10. Interactive Nigeria Map.....	8
11. UI/UX and Dashboard Design.....	9
12. Proposed Visualization Plan.....	9
13. Election Integrity / Anomaly Methodology.....	10
14. Data Sources and Provenance.....	10
15. Data Limitations and Assumptions.....	11
16. Technical Architecture.....	11
17. Development Plan.....	11
18. Final Deliverables.....	12
19. Skills Demonstrated.....	12
20. Success Criteria.....	13
21. Portfolio Positioning.....	13
22. Analytical and Ethical Principles.....	13
23. Conclusion.....	14
Appendix A — Project Scope at a Glance.....	15
Appendix B — Reference Basis.....	15

1. Executive Summary

The Voting & Election Performance Intelligence Dashboard is an interactive Excel-based Business Intelligence solution designed to analyse voter participation, election results and data-quality/integrity indicators across Nigeria's Federal and State elections conducted by the Independent National Electoral Commission (INEC).

The project has evolved from an initial voting dashboard concept into a question-driven analytical solution. Feedback from the Data Analyst community established two important design improvements: first, Election Type should be separated from Voting Method/Medium; second, the dashboard should be built around the three questions that matter most to an election audience: who is voting, who is winning, and whether the available data shows patterns that warrant further review.

INEC's stated constitutional functions include organising, undertaking and supervising elections for the President and Vice-President, Governors and Deputy Governors, members of the Senate, House of Representatives and State Houses of Assembly. INEC's official election-results portal also separates results into Presidential, Governorship, Senatorial, House of Representatives and House of Assembly elections. Citations will be finalised and added in the final portfolio documentation.

Important analytical boundary

The dashboard will not declare that an election is 'rigged' based solely on dashboard metrics. It will identify potential anomalies, inconsistencies and unusual patterns that may warrant investigation. Any conclusion about actual electoral fraud requires appropriate evidence, context and formal investigation.

2. Background and Rationale

Election data contains multiple dimensions: election cycle, office being contested, political party, candidate, geography, voter participation and vote outcomes. A conventional table can store these values, but it does not necessarily communicate the relationships between them efficiently.

This project therefore treats the dashboard as an analytical product rather than a collection of charts. The interface will be designed around the questions users are likely to ask and will guide them from high-level KPIs to geographical and constituency-level detail.

- Transform structured election records into actionable analytical information.
- Create an executive overview that communicates the most important results immediately.
- Use geographical visualization to reveal differences across Nigerian states.
- Enable users to move from national-level context to state, LGA, ward and polling-unit analysis where the dataset supports it.
- Use integrity indicators carefully and transparently rather than making unsupported claims.

3. Project Scope

3.1 Included Elections

The project will focus exclusively on elections conducted by INEC at the Federal and State levels. Local Government elections conducted by State Independent Electoral Commissions are explicitly excluded from the scope.

Government Level	Election Type	Included?
Federal	Presidential	Yes
Federal	Senatorial District	Yes
Federal	House of Representatives	Yes
State	Governorship	Yes
State	State House of Assembly	Yes
Local	Local Government Chairmanship	No
Local	Local Government Councillorship	No
FCT Local	Area Council elections	No — outside the agreed project scope

3.2 Historical Period

The core historical comparison will use the five most recent completed Nigerian general-election cycles: 2007, 2011, 2015, 2019 and 2023. The dataset will not imply that every election type occurred nationally on the same date or under identical conditions. Off-cycle elections and elections not held in a given state/year will be represented only where the project data explicitly supports them.

3.3 Geographic Scope

- Nigeria as the national level.
- 36 States and the Federal Capital Territory (FCT).
- State-level analysis.
- Local Government Area (LGA) as a geographic analytical dimension where available in the source data.
- Ward / Registration Area and Polling Unit where the dataset supports that granularity.
- Senatorial District for Senatorial elections.
- Federal Constituency for House of Representatives elections.
- State Constituency for State House of Assembly elections.

4. Project Objectives

1. Build an interactive Excel-based election intelligence dashboard.
2. Analyse voter registration, accreditation, participation and turnout.
3. Compare election performance across Federal and State election types.
4. Identify winning candidates and parties and quantify winning margins.
5. Analyse geographical voting patterns across Nigerian states and constituencies.
6. Provide an interactive Nigeria map that updates with relevant selections and metrics.
7. Develop KPI cards that communicate participation, results and integrity indicators.
8. Build an analytical framework for identifying potential anomalies and inconsistencies.
9. Apply UI/UX principles to create a clear, intuitive and executive-friendly dashboard.
10. Document the methodology, assumptions, limitations and findings as a professional portfolio case study.
11. Create a workbook architecture that can later be migrated or reproduced in Power BI.

5. Core Analytical Questions

5.1 Who Is Voting?

- How many voters are registered?
- How many voters are accredited?
- How many votes were cast?
- What is the turnout rate?
- Which states and LGAs show the highest and lowest participation?
- How does participation differ by election type?
- How does participation differ across election cycles?
- What participation patterns appear unusual and deserve review?

5.2 Who Is Winning?

- Which candidate received the highest number of valid votes?
- Which party performed best?
- What percentage of valid votes did the leading candidate receive?
- What is the winning margin?
- Where did the leading candidate or party perform strongest?
- How does candidate/party performance vary across states and constituencies?

5.3 Does the Data Show Potential Irregularity Indicators?

- Are votes cast consistent with accredited voters?
- Are registered, accredited, valid and rejected vote counts internally consistent?
- Are turnout rates unusually high or low relative to comparable areas?
- Are rejection rates unusually high?
- Are there locations with unusually concentrated candidate vote shares?
- Are there duplicate, missing or structurally inconsistent records?
- Which locations should be flagged for further review?

6. Election Taxonomy and Data Model

A central requirement of the redesigned project is to separate the office/election being contested from the method or medium through which voting is represented in the dataset.

Dimension	Examples	Purpose
Government Level	Federal / State	Groups the office by level of government
Election Type	Presidential / Governorship / Senatorial / House of Representatives / State House of Assembly	Identifies the office being contested
Election Cycle	2007 / 2011 / 2015 / 2019 / 2023	Supports historical comparison
Voting Method / Medium	Physical / Electronic scenario	Allows method-based analysis without confusing it with election type
Election Authority	INEC	Identifies the body responsible for the included election types
Geographic Level	State / LGA / Ward / Polling Unit	Supports geographical analysis
Constituency	Senatorial District / Federal Constituency / State Constituency	Matches the electoral structure

Voting Method clarification

Historical records will not be presented as evidence that Nigeria used electronic voting in the selected historical cycles unless the underlying source explicitly supports that claim. If electronic voting is included, it will be clearly labelled as a simulated/scenario dimension rather than a historical fact.

7. Dataset Architecture

The redesigned master dataset will be structured so that one analytical record can represent a candidate's result for a defined electoral unit.

Field Group	Representative Fields
Election	Election ID, Year, Cycle, Date, Government Level, Election Type
Geography	Region, State, LGA, Ward, Polling Unit
Constituency	Senatorial District, Federal Constituency, State Constituency
Political	Party, Party Code, Candidate
Participation	Registered Voters, Accredited Voters, Votes Cast
Results	Valid Votes, Rejected Votes, Candidate Votes
Calculated Metrics	Turnout %, Vote Share %, Rejection Rate %, Winning Margin
Integrity	Vote/Accreditation Check, Turnout Anomaly, Concentration Flag, Integrity Flag
Metadata	Source, Source Type, Data Status, Notes

8. Core Data Dictionary

Field	Definition	Type
Election ID	Unique identifier for an election context.	Text
Election Year	Year associated with the election cycle.	Integer
Government Level	Federal or State.	Category
Election Type	Office being contested.	Category
Voting Method	Physical or clearly labelled simulated electronic scenario.	Category
State	Nigerian state or FCT.	Category
LGA	Local Government Area associated with the electoral record.	Category
Ward	Registration Area/Ward where available.	Category
Polling Unit	Polling-unit identifier/name where available.	Text
Senatorial District	Senatorial district for Senate elections.	Category
Federal Constituency	Federal constituency for House of Representatives elections.	Category
State Constituency	State constituency for House of Assembly elections.	Category
Candidate	Candidate receiving votes.	Text
Party	Political party represented by candidate.	Category

Field	Definition	Type
Registered Voters	Number of registered voters in the relevant unit.	Integer
Accredited Voters	Number of voters accredited in the relevant unit.	Integer
Votes Cast	Total votes recorded as cast.	Integer
Valid Votes	Votes accepted as valid.	Integer
Rejected Votes	Votes rejected/invalid.	Integer
Candidate Votes	Votes received by the candidate.	Integer
Turnout %	Accredited or votes-cast participation relative to registered voters, per the defined methodology.	Percentage
Vote Share %	Candidate votes as a share of valid votes.	Percentage
Winning Margin	Difference between the winner and runner-up.	Integer
Integrity Flag	Analytical status based on defined consistency/anomaly rules.	Category

9. KPI Framework

KPI	Purpose
Registered Voters	Size of the registered electorate.
Accredited Voters	Participation through accreditation.
Voter Turnout %	Measures participation relative to the registered electorate.
Votes Cast	Total recorded votes.
Valid Votes	Accepted votes used for candidate results.
Rejected Votes	Votes rejected as invalid.
Rejection Rate %	Rejected votes relative to votes cast.
Leading Candidate	Identifies the candidate with the highest valid vote total.
Winning Vote Share %	Shows the winner's share of valid votes.
Winning Margin	Shows the gap between first and second place.
Flagged Locations	Counts electoral units requiring analytical review.

10. Interactive Nigeria Map

The dashboard will include an interactive Nigeria state map as a core analytical visualization. The map is justified because geography is fundamental to understanding voter participation and election outcomes.

- Display all 36 states and FCT at the national overview level.
- Apply a visual scale to selected metrics such as turnout, vote share, winning margin or flagged locations.
- Respond to relevant dashboard filters.
- Highlight a selected state while preserving national geographic context.
- Support state-level comparison and data storytelling.
- Optionally expose a contextual detail panel for the selected state.

Map principle

The map will be an analytical component, not decoration. Every map metric will correspond to a defined business question or KPI.

11. UI/UX and Dashboard Design

The dashboard will adopt an application-style layout inspired by modern executive BI interfaces rather than the horizontal worksheet structure used in the earlier Transportation Performance Dashboard.

- Permanent left navigation sidebar.
- Clear executive title and subtitle.
- Filters grouped at the top of the analytical workspace.
- Large KPI cards for immediate comprehension.
- Question-driven chart sections.
- Dedicated Key Insights panel.
- Consistent typography, spacing, icons and card design.
- Minimal visual clutter and strong information hierarchy.
- Consistent semantic use of colour.

Proposed Navigation

Page	Primary Question
Executive Overview	What is happening overall?
Voter Participation	Who is voting?
Election Results	Who is winning?
Election Integrity	What potential irregularity indicators exist?
Calculations	How are KPIs and analytical indicators derived?

Page	Primary Question
Data	What is the underlying structured dataset?

12. Proposed Visualization Plan

Area	Recommended Visuals
Executive KPIs	KPI cards with current context and concise definitions
Geography	Interactive Nigeria state map
Participation	Registered vs accredited, turnout ranking, participation trend
Results	Candidate ranking, vote share, winning margin, party performance
Integrity	Consistency indicators, anomaly ranking, rejection-rate analysis
Insights	Dynamic text-based insight panel

13. Election Integrity / Anomaly Methodology

The integrity component will use transparent, rule-based analytical checks. Thresholds will be documented and treated as portfolio methodology rather than official electoral standards.

Indicator	Illustrative Analytical Check	Output
Vote/Accreditation Consistency	Votes cast should not exceed accredited voters under the selected data definition.	Pass / Review
Participation Consistency	Compare registered, accredited and votes-cast fields for logical consistency.	Pass / Review
Turnout Anomaly	Flag unusually high/low turnout relative to a defined comparison group.	Normal / Review
Rejection Rate	Flag unusually high rejected-vote rates.	Normal / Review
Vote Concentration	Identify unusually concentrated candidate vote shares.	Normal / Review
Duplicate Records	Detect duplicate analytical records where unique keys should be unique.	Clear / Review
Missing/Invalid Values	Identify missing or structurally invalid records.	Clear / Review

Interpretation rule

A flagged indicator means 'requires further review' — it does not mean 'fraud confirmed'. This distinction is a deliberate part of the project's analytical ethics.

14. Data Sources and Provenance

Where official historical election data is used, the project will prioritize INEC publications and election-result materials. INEC's official results portal provides separate result categories for Presidential, Governorship, Senatorial, House of Representatives and House of Assembly elections.

- Official INEC election-result publications and downloadable materials where available.
- Official INEC election guidelines, reports and electoral administration documentation.
- Clearly labelled synthetic or simulated data where official records are unavailable at the required analytical granularity.
- Source and source-status fields retained in the dataset so that historical facts are not confused with portfolio-generated values.

For the final portfolio report, source URLs and access dates will be documented alongside the relevant datasets and methodology.

15. Data Limitations and Assumptions

- Historical election data is not guaranteed to be available at identical granularity across all five election cycles.
- Some constituency boundaries and administrative structures can change over time; the dataset must preserve the geography appropriate to the relevant election year where possible.
- Off-cycle elections may not align with the general-election cycle.
- If synthetic data is used to complete a portfolio demonstration, it will be explicitly labelled as synthetic.
- Electronic voting should not be represented as a historical Nigerian election method unless supported by the underlying source.
- Anomaly indicators are analytical signals and not legal determinations of electoral malpractice.

16. Technical Architecture

The workbook will follow a layered architecture so that the analytical engine remains separate from presentation.

Data Layer → Calculation Layer → Pivot/Analysis Layer → Visualization Layer → Interaction/Automation Layer

- Data sheet for structured records.
- Calculation sheet for derived fields and KPI logic.
- PivotTables/PivotCharts for aggregation.
- Slicers for interactive filtering.
- Dashboard sheets for presentation.
- VBA for navigation, dynamic interface behaviour and selected automation.
- Named ranges and helper cells where required for stable dashboard references.

17. Development Plan

12. Lock final project requirements and proposal.
13. Design the complete dataset architecture and data dictionary.
14. Identify and document official data sources.
15. Prepare the historical/synthetic portfolio dataset.
16. Perform data-quality checks.
17. Build calculation and KPI engine.
18. Build PivotTables and analytical views.
19. Design the dashboard wireframe before visual construction.
20. Develop Executive Overview.
21. Develop Voter Participation page.
22. Develop Election Results page.
23. Develop Election Integrity page.
24. Develop interactive Nigeria map.
25. Add slicers and cross-filtering.
26. Add VBA navigation/automation.
27. Test calculations, filters, map behaviour and visual consistency.
28. Write final insights and project report.
29. Prepare GitHub, Google Drive and LinkedIn portfolio assets.

18. Final Deliverables

- Interactive Excel Voting & Election Performance Intelligence Dashboard (.xlsm).
- Final Project Proposal.
- Final Project Report.
- Project Summary.
- Data Dictionary.
- Dataset / source documentation.
- Dashboard screenshots.
- GitHub repository.
- Portfolio documentation.
- LinkedIn project showcase.

19. Skills Demonstrated

Skill	Skill
Data Analysis	Business Analysis
Business Intelligence	Operational / Performance Analysis
KPI Development	Data Validation
Data Visualization	Dashboard Development
PivotTables and PivotCharts	Advanced Excel Formulas
Interactive Slicers	VBA Automation
Anomaly Detection	Geospatial Data Visualization
Data Storytelling	UI/UX-informed Dashboard Design
Executive Reporting	Data-driven Decision Support

20. Success Criteria

The project will be considered successful when the dashboard allows a user to:

- Understand the overall election context within seconds of opening the dashboard.
- Identify who is voting and where participation is strongest/weakest.
- Identify who is winning, by how much and where.
- Explore election performance across Federal and State election types.
- Use filters to change the analytical context dynamically.
- Use the Nigeria map to understand geographical differences.
- Identify locations or patterns flagged for further analytical review.
- Understand what each KPI and integrity indicator means.
- Trace key results back to a transparent calculation/data layer.

21. Portfolio Positioning

This project is intended to demonstrate progression from practical Excel reporting into more advanced Business Intelligence and data-product thinking.

Portfolio Project	Primary Capability Demonstrated
Agent Performance KPI Dashboard	Operational KPI reporting and performance monitoring
Transportation Performance Dashboard	Executive BI, slicers, MoM analysis and VBA automation
Voting & Election Performance Intelligence Dashboard	Multi-dimensional analytics, anomaly analysis, geospatial visualization and UI/UX-informed dashboard design

The workbook will also be structured with future Power BI migration in mind. Dimensions such as Election Year, Election Type, State, LGA, Constituency, Party and Candidate will be kept consistent so the analytical concepts can later be reproduced in Power BI.

22. Analytical and Ethical Principles

- Do not present simulated values as official historical results.
- Do not make unsupported claims about election fraud.
- Show methodology behind calculated indicators.
- Preserve source provenance where possible.
- Use clear labels for assumptions and synthetic data.
- Prefer transparent indicators over opaque scoring.
- Design visualizations to inform rather than sensationalize.

23. Conclusion

The Voting & Election Performance Intelligence Dashboard will combine practical Excel skills with a more mature Business Intelligence approach. Its purpose is not merely to reproduce election results, but to help users understand participation, results, geography and potential data irregularities through an interactive and visually coherent analytical environment.

The final solution will be built around the principle:

DATA → ANALYSIS → VISUALIZATION → INTERACTION → INSIGHT → DECISION SUPPORT

Appendix A — Project Scope at a Glance

Component	Final Decision
Primary Tool	Microsoft Excel
Dashboard Style	Application-style / UI/UX-informed
Navigation	Left sidebar
Map	Interactive Nigeria state map
Election Level	Federal + State only
Election Types	Presidential, Governorship, Senatorial, House of Representatives, State House of Assembly
Local Government Elections	Excluded
Historical Cycles	2007, 2011, 2015, 2019, 2023
Core Questions	Who is voting? Who is winning? Potential irregularity indicators?
Integrity Output	Normal / Review / Potential Anomaly — not a fraud verdict
Main Filters	Election Year, Government Level, Election Type, State, LGA, Constituency, Party, Candidate
Portfolio Goal	Demonstrate Excel BI, data analysis, visualization, anomaly analysis and UI/UX thinking

Appendix B — Reference Basis

The project scope was aligned with publicly available INEC descriptions of the Commission's electoral responsibilities and official election-result categories. INEC states that it organises, undertakes and supervises elections for the President and Vice-President, Governors and Deputy Governors, the Senate, House of Representatives and State Houses of Assembly. INEC's official results portal separately lists Presidential, Governorship, Senatorial District, House of Representatives and House of Assembly results. The final portfolio report should include the exact source URLs used for any historical dataset.